



DPUDemultiplexer

1. Inputs:

Switch

Switches the input to one of the outputs.

Default 0

SwitchOffset

Offset which is added to *Switch*.

Default 0

In

Input that is switched to the outputs.

Multiplier[1...20]

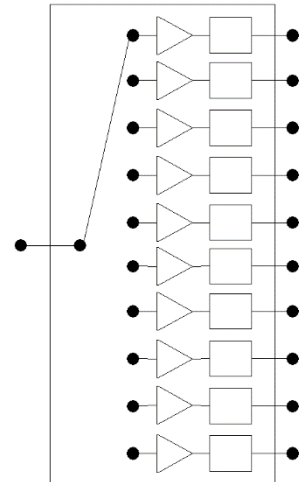
Multiplier for outputs.

Divisor[1...20]

Divisor for outputs.

Offset[1...20]

Offset for the different outputs.



2. Dynamic parameters:

OutHold[1...20]

= 1: Output holds its last value, even if it isn't switched to input anymore.

= 0: Output is set to its default value, when it isn't switched to input anymore. (default behaviour).

OutMin[1...20]

Minimum value for corresponding output.

OutMax[1...20]

Maximum value for corresponding output.

Default[1...20]

Default value for every output. Output *i* is set to this default value when it isn't switched to the input and if *Out-Hold*_{*i*} = 0.

3. Outputs:

Out[1...20]

Output 1 to 20.

4. Functional principle:

The DPUDemultiplexer connects the input *In* with the outputs *Out1* to *Out10* dependent on the inputs *Switch* and *SwitchOffset*. For example, if the input *Switch* is set to the value 1 (and *SwitchOffset* to 0), then input *In* is connected to the output *Out1*. In general, the output *<i>* is connected to the input *In* if *Switch* + *SwitchOffset* = *<i>*.

The final value of output *OutX* is calculated with the following formula:

$$OutX = \frac{In \cdot MultiplierX}{DivisorX} + OffsetX$$

By using parameter *OutHoldX* it can be determined whether the output *OutX* falls back to value *DefaultX* when it is no longer connected to the input, or if the last displayed value is held.



5. Example:

Depending on the *Switch* input exactly one (or none if *Switch* = 0) HMI will be shown.

```
DPUDemultiplexer DisplaySwitch
{
    Computer = {DSP};
    Index = 123;
    In = 1;
};
Connections = {
    DisplaySwitch.Out1 -> ClusterDisplay.Show,
    DisplaySwitch.Out2 -> CenterDisplay.Show,
    DisplaySwitch.Out3 -> Headup.Show
};
```